

Learning to be Outside: How Setting Up Girls' Schools in Late Qing and Minguo Affects Gender Norms and Labor Supply

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Abstract

How will girls' education influence the gender norms and labor supply? China in the late Qing and Minguo witnessed a dramatic increase in the number of girls' schools, followed by an increase in female labor participation rate in all occupations. Extracting data from *Minguo Research on Education*, *Minguo Research on Economic Situation*, this paper analyzes how the increase of girls' schools and girl students influences the gender norms and female labor supply at that time. It turned out that 1,000 more girl students will lead to a 3-point decrease in sex ratio, and 1 percent increase for girl students over the female population will make the estimated Female labor participation rate double. These results were robust after detrending over time and by using different signals for girls' education.

Key Words: Girls' Schools, Gender Norms, Female Labor Participation Rate, Sex Ratio, Late Qing and Minguo

1 Introduction

Education has long been regarded as an effective way to improve gender equality both in the sense of women empowerment and related social norms. [Unterhalter et al. \(2014\)](#) and [Ramachandran \(2009\)](#) have comprehensively discussed the correlation between education and gender equality. Moreover, there are also papers focusing on the effect of improving girls' education through more investments on gender equality. ([Aikman and Rao, 2012](#); [Ember, 2020](#)) All these works have shown that improving education conditions for girls is positively correlated with gender equality and female labor force participation, thus having a better economic outcome. ([Kabeer and Natali, 2013](#)) This paper analyzes similar patterns of this but uses a special period in late Qing Dynasty where there are some girls' schools, thus testing the effect of increasing number of girls' schools on gender norms and female labor participation at that time.

This paper studies the emergence of girls' schools. Before Late Qing Dynasty, girls and women were mainly not allowed to step outside their houses when they were young, and most girls would not go through typical education but only learns some sewing and cooking stuff with only a few exceptions. ([Hinsch, 2018](#)) Then, this was firstly changed by missionaries who came to China, and later on there were more and more schools with different functions. This incentivized more and more girls to get educated and then participated in the labor force and political issues.

Due to data scarcity at that time and there weren't many papers focusing on girls' schools at that time, I collected data from *Minguo Research on Education* and *Minguo Research on Economic Situation* and tried to summarize them into a table on a province level. Due to data availability, I performed Ordinary Least Squares with this cross-sectional data. Meanwhile, I also used education situations and female labor participation rate(below FLPR) on a country level each year for further analysis. Then, I found that increasing number of girls' schools and girls' students led to fewer discrimination against girls and women and higher female labor participation rate. This result holds even controlling for overall number of schools and some other factors that may influence the gender norms and FLPR. And there shouldn't be a reversed cause relationship since in late Qing, nearly no women worked outside families. (Shi, 2018)

Then, this paper tries to analyze the mechanism within this result. That is, how launching more girls' schools resulted in households' willingness to send their daughters to schools. From the side of poor families, since at the beginning girls' schools with religious purpose (for example, girls' schools created by Catholic missionaries) would provide meal and accommodation for girls and even give them some money (Wiest, 2011), poor families could save raising costs for their girls. While for rich families where raising costs were not a big deal, the prevalence of girls' schools internalizes the norms cost. In the culture of ancient China as a agricultural society, the norm of preferring boys to girls was so important that modern families trying to educate their girls would face pressure from the society. But the emergence of girls' schools was making girl education a normal thing, which internalizes the norms cost, similar to the mechanism mentioned in Field et al.(2021).

This paper contributes to the discussion of girls' education on improving gender equality in three ways. Firstly, it compiled data in education from 1905 to 1925 into digital table-form, and summarized data from several papers.(Mi, 2014; Wiest, 2011; Pan, 2001) Secondly, since there is a strong causal relationship both from girls' education to FLPR and from FLPR to girls' education, (Goldin, 1984; Goldin, 1986) this paper identifies the effect of girls' education on FLPR using a special period in China. Thirdly, it proposes different mechanisms for poor families rich families and internalizes the norms cost.

2 Historical Background

2.1 Education of Girls

In ancient China, girls and women were generally regarded as attachments to men and were not allowed generally to learn reading and writing.(Bernhardt, 1999) The beginning of girls' education with literacy skills was from the outside world, where missionaries from the western world tried to preach the religious beliefs. And the way many religions including Catholic was through education and schools. (Wiest, 2011) So, the first girl school in China was set up by Aldersoy who is a missionary from the U.K. at that time.

However, the real emergence of girl schools was at the end of 19th century and at the beginning of 20th century, when the Qing Dynasty announced officially some regulations that allowed schooling of girls including *Women's Normal School Constitution* and *Women's Elementary School Constitution*. (Qu, 2012) Afterwards, the number of girls' schools and girls' students increased dramatically. Also, the type of girl schools and girl students also ranged from primary, middle, secondary, normal and technical. For example, in the May of 1898, the first girl school created by Chinese people was launched. And in 1913, the first

Year	Fraction(%)
1915	37.8
1916	41.7
1917	42.8
1918	37.1
1919	44.7
1920	40.5

Table 1: The Fraction of Female Workers of Total Workers in 1915-1920.

girl college, Ginlin College, was founded. Using data for girl education, we found that the number of girl students increased dramatically from 1905 to 1925. (Figure 1)

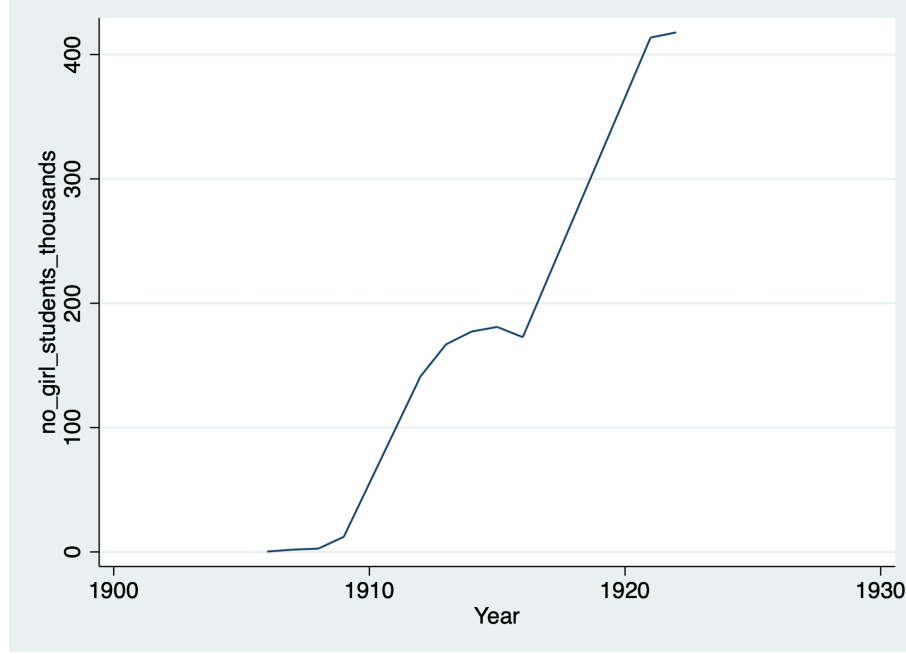


Figure 1: Number of girl students in China from 1905 to 1925. (in thousands)

Source: Author draw from data

2.2 Norms of Women and Labor Participation

Inheriting from the agricultural society, girls and women were valued much less than boys and men. (Alesina, Giuliano, and Nunn, 2013) Specifically, as for labor participation for men and women, men were supposed to learn and work "outside" while women were supposed to learn and work "inside", thus making FLPR in Qing Dynasty and before to be really low since home production was not regarded as work and people working at home were not considered labor force.

However, after the emergence of girl education in various forms, along with 1911 Revolution and 1919 May Fourth Movement, female stepped outside the house and participated in the labor force as workers, political leaders and many other occupations. Goodman and Larson (2005) shows that in the whole country, female workers have already been critical as a fraction of total workers. (Table 1)

3 Data

3.1 Girls' Education

During Late Qing and Minguo, there have been data on the education situations both of the whole country and of different provinces. By extracting key information about girl schools and girl students, we can firstly figure out that girl education differed quite a lot in different provinces. (Figure 2 and 3)

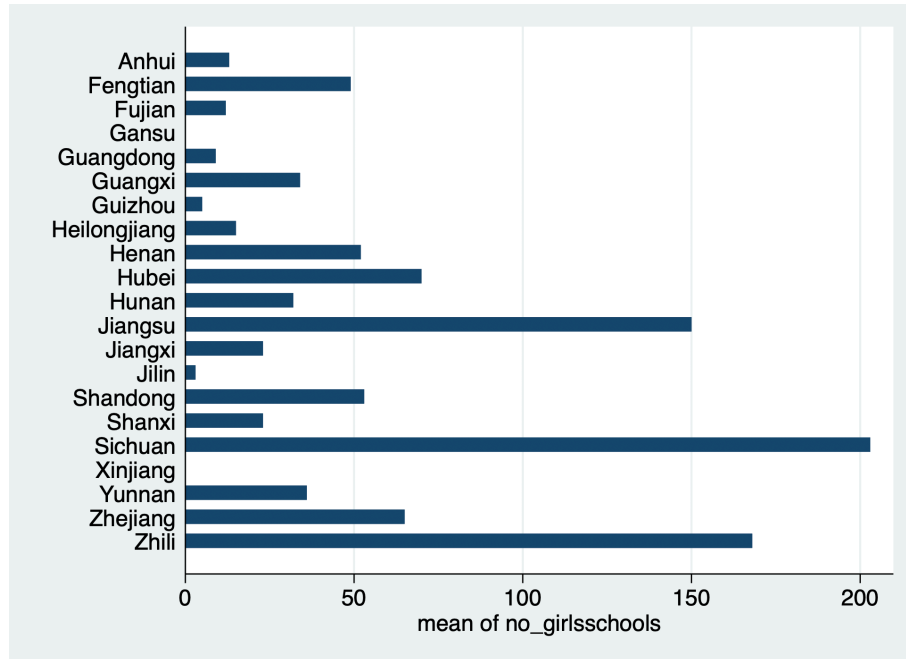


Figure 2: Number of girl schools in different provinces of China in 1909.
Source: Author draw from data

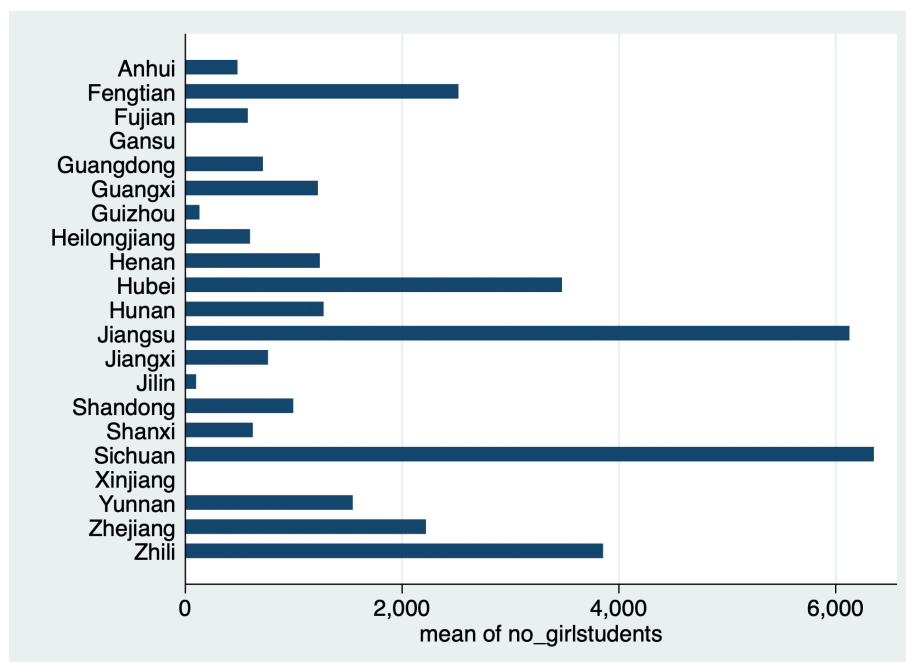


Figure 3: Number of girl students in different provinces of China in 1909.
Source: Author draw from data

Then, since different provinces have quite different total population and households, with data from the

1909 census, it's shown that girl student ratio was still quite low compared with total number of students. And ratio of girl students of female population still lagged far behind that of male. Moreover, there's no direct link of the number of girl students with the ratio of girl students over the total population. (Figure 4, 5)

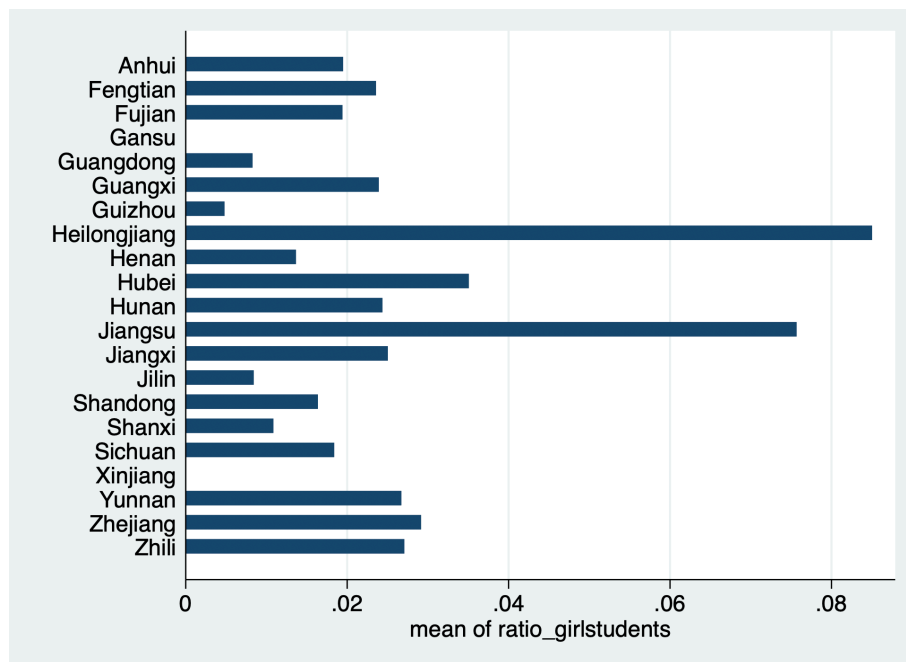


Figure 4: Ratio of girl students over total students in different provinces of China in 1909.
Source: Author draw from data

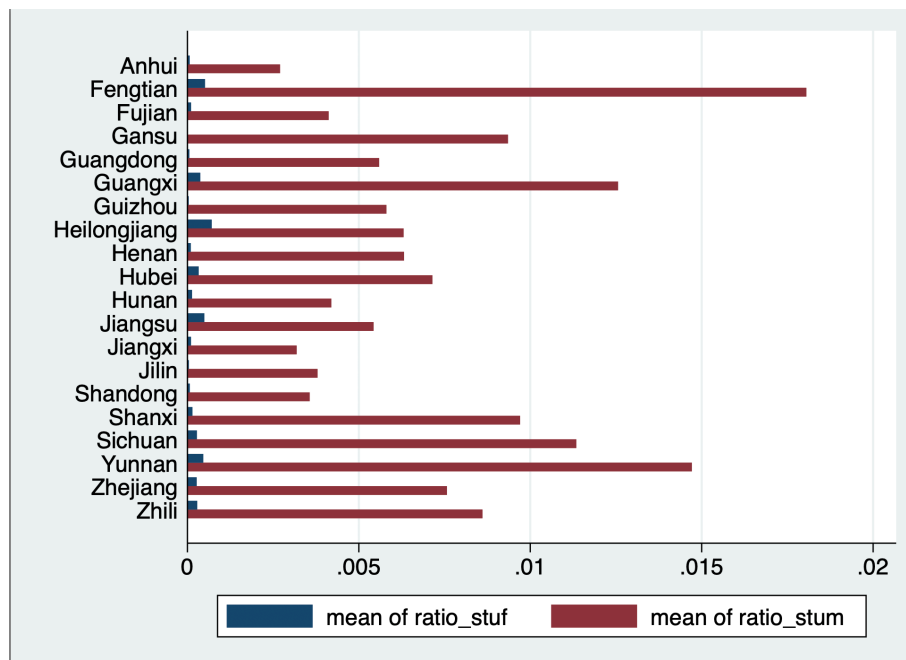


Figure 5: Ratio of students over corresponding total population in different provinces of China in 1909.
Source: Author draw from data

3.2 Gender Norms and Labor Participation

In this paper, one key thing is to identify the gender norms and FLPR at that time since both couldn't be got directly from the existing data.

As for gender norms, this paper seized to use the sex ratio, which was collected in two census (Mi, 2014), as a representation for people's preference for girls. This was related to the concept of excess female mortality, which was frequently used as a index for gender equality. (Coale, 1991; Hausmann, Tyson, Zahidi, et al., 2012) In China's modern history, the change of sex ratio can be viewed as an approximate for excess female mortality since the preference for boys would force girls to live in a poor situation or even be abandoned, which will be represented in the sex ratio.

And for labor participation, since in most sources of information FLPR was not noted or even mentioned before 1949, this paper turns to number of news concerning female labor as an approximate to the real FLPR. And the database is Shanghai Library, where there was a search engine for all the passages and reports emerged in newspapers, journals and books in a certain period of time. ([Search Engine for news papers and journals in Shanghai Library](#))

To ensure that the number of news about female labor was a good estimator for the real FLPR, I also tried to use the number of news about girl schools to estimate the number of girl schools and found the average error is within 5% (This result and detailed data package can be given upon notification), thus making the number of news concerning female labor a promising estimator for the FLPR.

Also, the number of news about female increased drastically from 1911 and peaked in 1936, which could be explained by the May Fourth Movement and the increasing prevalence of new ideas about males and females.(Figure 6)

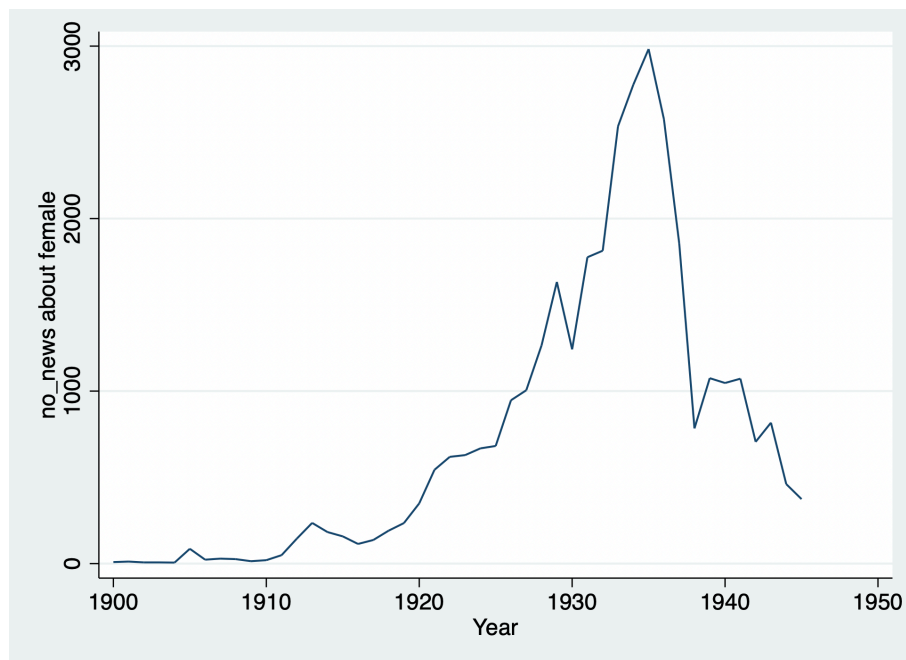


Figure 6: Number of news concerning females in most newspapers and journals, 1900-1945.
Source: Author draw from data

4 Empirical Strategy and Result

The key question here is whether increasing number of girl schools or students and the increasing ratio would lead to lower excess female mortality, represented by the sex ratio in different provinces. And we also try to control the region specific characteristics like the population and households. And we try to control the economic development of each province using the ratio of total students over the whole population since they are positively correlated. (Lewis, 1962)

Thus, we specify the estimation equation to be:

$$y_i = \beta_0 + \beta_1 GEdu_i + \beta_2 RBS_i + X'\delta + u_i$$

where y_i is the sex ratio in each province or the number of news. GEDu is represented by the number of girl students, ratio of girl students and ratio of girl schools separately. RBS is the ratio of boy students over the whole male population. And we have X as a set of vectors controlling for region-specific factors like the total population and the economic wellbeing.

4.1 Gender Norms

Here is the result for the effect of girl education on gender norms, where y represents sex ratio in Figure 7. We can see that increasing the number of girl students, ratio of girl students and ratio of girl schools will significantly decrease the number of sex ratio, which proved to improve gender equality since the sex ratio for all provinces were all greater than 105, which was the natural ratio for human beings.

VARIABLES	(1) y	(2) y	(3) y
no_girlstudents	-0.003** (-2.15)		
no_households	0.000 (0.31)	0.000 (0.42)	0.000 (0.02)
no_population	-0.000 (-0.03)	-0.000 (-0.76)	-0.000 (-0.12)
ratio_stum	-25,355.735*** (-5.30)	-31,653.389*** (-12.77)	-28,150.840*** (-7.17)
ratio_stutotal	46,069.833*** (5.24)	56,591.469*** (12.63)	50,360.644*** (7.08)
ratio_girlstudents		-241.833*** (-8.13)	
ratio_girlschool			-178.078*** (-3.91)
Constant	118.465*** (29.44)	128.932*** (81.00)	126.205*** (50.47)
Observations	20	20	20
R-squared	0.682	0.926	0.797

t-statistics in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Figure 7: Effect of Girl Education on Gender Norms

An increase of 1000 girl students will drag down the sex ratio by 3, which plays an important role in

improving the gender norms or discrimination among people.

4.2 Labor Participation

Now let's turn to the effect of girl education on FLPR, where y represents the number of news concerning female labor in Figure 8. From the first column we can derive that increase of girl students lead to higher FLPR, and this effect is robust when we control for the time trend in column 2. Also, using the ratio of girl students over the female population as the girls' education will give a similar result.

VARIABLES	(1) y	(2) y	(3) y
no_girlsstudentse(except religious)	0.0000843*** (6.92)	0.000** (3.35)	
no_students	-0.00000*** (-5.13)	-0.000*** (-5.10)	-0.000*** (-5.17)
Year		1.708 (1.67)	3.288*** (3.99)
no_girlstudents(percentage)			241.198** (2.67)
Constant	6.804** (3.25)	-3,248.492 (-1.67)	-6,262.175*** (-3.99)
Observations	11	11	11
R-squared	0.890	0.922	0.899

t-statistics in parentheses
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Figure 8: Effect of Girl Education on FLPR

5 Mechanism

After showing that increasing number of girl schools, girl students and relevant ratio will shift people's gender norms and lead more women to participate in the labor force in many aspects. The remaining question would be why households would be willing to send their daughters to those schools. Here, I will divide the mechanism into two parts.

For poor households, raising a girl could be expensive and consumes a lot of food at home. And in the folk or feudal society, when girls get married they no longer belong to this family, which makes this "investment" be valueless in the long run. While some schools especially religious ones provide free meals and accommodation for girls attending that school to attract more people, thus those poor families would be willing to send their daughters to these schools mainly for saving money in the beginning.

For rich households, things become different since they could usually afford the food costs and with some marriages, they could be friends with other rich families, which proves to be beneficial in the long run. Then, the setting up of girls' schools mainly internalizes the norms costs. Norms costs refer to the social pressure the family has to endure for sending their daughter to schools if the society is against this idea. But the introduction of girls' schools and relevant policies relieve this constraints, thus making more rich families send their daughters to schools after weighing the costs and benefits. (For a more detailed

analysis, see [Field et al., 2021](#))

6 Discussion and Future Work

This paper now could serve as a preliminary result of the girls' education on gender equality and FLPR during the period of Late Qing Dynasty and Minguo. This period is special for identifying this causal effect since the female labor participation rate as the beginning is quite low. And for further analysis, several paths can be followed.

Firstly, since in the *Minguo Research on Education* and *Minguo Research on Economic Situation* there were detailed data for each province and detailed information for primary, middle and secondary schools, a county-based database could be constructed, making the sample size to be larger.

Secondly, since religious girls' schools, state-owned girls' schools and private girls' schools provide different courses and teach different skills, if these girls' schools could be analyzed separately, more interesting results could be found.

Thirdly, the number of female teachers in girls' schools could be another interesting thing to focus on. These group of people will both influence the FLPR and the development of education. ([Cappelli and Quiroga Valle, 2021](#)) Thus, getting data of the number of female teachers in different years in different provinces can help to explain FLPR and girls' education better.

7 Conclusion

This paper tests how setting up more girls' schools and increasing number of girl students will influence the gender norms toward equality and the FLPR. By using data extracted from original statistics in Late Qing and Minguo, I have shown that better girls' education will lead to lower sex ratio and lower excess female mortality, and also more news concerning female labor, which proves to be a good estimator for FLPR. Furthermore, it tries to analyze how people of different income groups respond to the setting up of more girls' schools.

The emergence of girls' schools helped the gender equality, and thus promoting boys and girls to learn in the same classroom. This has implication even for today's society. The existence of gender inequality inherited from history demonstrates the significance of the practice of women leaders' or R&D program like [Next-Generation-Women-Leaders from Mckinsey](#). It promotes gender equality and abandons these programs. This path echoes with the education development during Late Qing and Minguo in China.

8 Appendix

Here I provide the picture of the refined data collected from the *Minguo Research on Education*, *Minguo Research on Economic Situation* and some research papers, and the original format could be provided upon notification.

Region	Region	ENG	Region no	no	population no	households	sex ratio	no	fpopulation no	mpopulation no	schools	no	students	no	girls	school no	girl students	ratio	girl school/ratio	girl student ratio	stun	ratio	stuf	ratio	stutotal
直隸	Zhili	1	2950773	5721623	119.8	13423919	16081854	2201	142247		168	3853	7.63%	2.71%	0.8606%	0.0287%	0.4821%								
奉天	Fengtian	2	10090604	1650751	117.6	4013443	5760261	2708	108607		49	2159	1.81%	2.38%	1.8052%	0.0512%	0.9991%								
吉林	Jilin	3	2393744	729661	122.0	2224800	3060854	338	11785		3	99	0.89%	0.84%	0.3795%	0.0043%	0.2178%								
黑龙江	Heilongjiang	4	1858792	245957	120.8	841844	1016048	196	7009		15	596	7.65%	8.50%	0.6306%	0.0708%	0.3771%								
山东	Shandong	5	31047430	5380277	117.3	14287819	16759611	4396	60765		55	995	1.21%	1.84%	0.3566%	0.0070%	0.1957%								
山西	Shansi	6	10140469	2001776	135.5	4200753	5839716	2333	57291		23	622	0.99%	1.09%	0.9704%	0.0144%	0.5645%								
陕西	Shanxi	7	8071792	1405342	120.0	3668996	4402796	2953	59196																
河南	Henan	8	26849405	4661566	111.5	12716296	14776059	3773	90824		52	1240	1.38%	1.37%	0.6318%	0.0009%	0.3377%								
江苏	Jiangsu	9	26166138	5384227	111.5	12381739	13752468	2462	80947		100	6126	6.09%	7.57%	0.5429%	0.0495%	0.3084%								
安徽	Anhui	10	16220052	3141184	123.1	7274340	8954712	865	24674		13	481	1.50%	1.95%	0.2702%	0.0066%	0.1520%								
浙江	Zhejiang	11	17997058	3891277	118.5	8237006	9760852	2165	70114		65	2119	3.00%	2.92%	0.7571%	0.0269%	0.4229%								
江西	Jiangxi	12	16725085	3679863	125.6	7617867	9311811	1262	30428		23	762	1.82%	2.50%	0.3186%	0.0103%	0.1819%								
湖北	Hubei	13	23941688	4038625	126.6	10565617	13176071	2886	90664		70	3474	2.43%	3.51%	0.7146%	0.0329%	0.4138%								
湖南	Hunan	14	23712203	4281664	127.3	9552223	12139989	1457	52329		32	1275	2.23%	2.44%	0.4139%	0.0133%	0.2410%								
四川	Sichuan	15	5209128	9190177	130.1	22972462	29004866	10661	345383		203	6352	1.90%	1.84%	1.1344%	0.0277%	0.6534%								
广东	Guangdong	16	28058658	5052418	120.5	12722615	15333643	1794	86437		9	715	0.50%	0.83%	0.5590%	0.0056%	0.3081%								
广西	Guangxi	17	7239437	1381372	121.5	3268369	3971968	1328	51697		34	1222	2.56%	2.39%	1.2560%	0.0374%	0.7058%								
云南	Yunnan	18	7174884	1501861	114.1	3351184	3823700	1944	57008		36	1544	1.85%	2.47%	1.4715%	0.0461%	0.8057%								
贵州	Guizhou	19	8503054	1771533	119.9	3866782	4636272	1811	27036		5	130	0.28%	0.48%	0.5803%	0.0034%	0.3180%								
福建	Fujian	20	12314842	2379139	128.9	5476121	7095721	678	29653		12	576	1.77%	1.94%	0.4119%	0.0105%	0.2366%								
甘肃	Gansu	21	4699494	908460	109.8	2239988	2459306	1243	22996		0	0	0.00%	0.00%	0.9350%	0.0000%	0.4893%								
新疆	Xinjiang	22	2149654	465880				462	6910		0	0	0	0.00%											

Figure 9: Detailed Census and Education Information for Different Provinces in 1909
Source: Author extract from books and papers

Year	no	girls	students	no	students	no	news	about female labor	no	news	about female	no	news	about female education	no	girl students	percentage
1900							0			9		0				#DIV/0!	
1901							0			12		0				#DIV/0!	
1902							0			7		0				#DIV/0!	
1903							0			7		0				#DIV/0!	
1904							0			6		0				#DIV/0!	
1905							0			86		2				#DIV/0!	
1906	306		468220				0			23		8				0.000653539	
1907	1853		883218				0			29		0				0.00209801	
1908	2679		1144209				0			26		1				0.002341355	
1909	12164		1536909				0			14		0				0.007914587	
1910							0			20		3				#DIV/0!	
1911							0			49		0				#DIV/0!	
1912	141130		2933387				3			145		1				0.04811162	
1913	166964		3643206				2			236		1				0.045828866	
1914	177273		4075338				1			183		0				0.043498969	
1915	180949		4294251				0			158		0				0.0421375	
1916	172724		3974454				1			114		0				0.043458548	
1917							0			137		1				#DIV/0!	
1918							2			191		3				#DIV/0!	
1919							5			235		0				#DIV/0!	
1920							8			349		0				#DIV/0!	
1921	413679		4987647				25			544		2				0.082940713	
1922	417820		6615772				6			619		4				0.063155139	
1923							10			629		8				#DIV/0!	
1924							12			668		8				#DIV/0!	
1925							4			682		1				#DIV/0!	
1926							10			947		6				#DIV/0!	
1927							7			1005		4				#DIV/0!	
1928							5			1265		3				#DIV/0!	
1929							15			1633		2				#DIV/0!	
1930							13			1242		6				#DIV/0!	
1931							12			1776		1				#DIV/0!	
1932							6			1814		4				#DIV/0!	
1933							21			2535		9				#DIV/0!	
1934							23			2776		3				#DIV/0!	
1935							21			2983		8				#DIV/0!	
1936							18			2577		4				#DIV/0!	
1937							21			1858		4				#DIV/0!	
1938							4			784		1				#DIV/0!	
1939							6			1075		1				#DIV/0!	
1940							11			1047		3				#DIV/0!	
1941							10			1072		1				#DIV/0!	
1942							10			706		1				#DIV/0!	
1943							29			817		1				#DIV/0!	
1944							6			461		0				#DIV/0!	
1945							1			374		0				#DIV/0!	

Figure 10: News concerning female labor and Education 1900-1945
Source: Author extract from books and papers

Year	Region	Region_no	Sex Ratio	no_girls_schools_t-2	no_girls_students_t-2	no_schools_t-2	no_students_t-2	no_girls_schools_t-1	no_girls_students_t-1
1909	直隶	1	119.8	121	2523	9377	181691	179	4443
1909	奉天	2	117.6	12	694	1352	51579	5	229
1909	吉林	3	132	0	0	38	2713	1	46
1909	黑龙江	4	120.8	2	90	77	4152	10	374
1909	山东	5	117.3	1	54	3905	55046	47	993
1909	山西	6	135.5	5	149	1055	28746	15	478
1909	陕西	7	120	10	154	2043	43757	10	154
1909	河南	8	111.5	3	84	1685	40077	50	1184
1909	江苏	9	111.3	96	4198	1616	57329	106	4509
1909	安徽	10	123.1	2	86	293	8606	10	271
1909	浙江	11	118.5	32	995	446	45537	56	1598
1909	江西	12	125.6	6	155	456	15134	13	547
1909	湖北	13	126	12	477	1504	58742	13	1481
1909	湖南	14	127.3	7	412	728	30003	16	935
1909	四川	15	130.1	70	2246	15943	223617	159	4738
1909	广东	16		6	391		75733	6	539
1909	广西	17		17	589	502	22277	18	760
1909	云南	18	114.1	18	1027	880	22957	7	147
1909	贵州	19	119.9	5	267	305	13635	2	80
1909	福建	20	128.9	3	244	385	22029	5	141
1909	甘肃	21	109.8	0	0	165	10961	1	11
1909	新疆	22	110.1	0	0	102	1119	7	64
1912	直隶	1	123.2	168	3853	2201	142247		
1912	奉天	2	123.7	49	2519	2708	106867		
1912	吉林	3	131	3	99	338	11745		
1912	黑龙江	4	140.8	15	596	196	7009		
1912	山东	5	107.2	53	995	4396	60765		
1912	山西	6	132.4	23	622	2333	57291		
1912	陕西	7	161.6	15	212	2953	59196		
1912	河南	8	119.5	52	1240	3773	90824		
1912	江苏	9	109.1	150	6126	2462	80947		
1912	安徽	10	122.7	13	481	865	24674		
1912	浙江	11	117.1	65	2219	2165	76114		
1912	江西	12	131	23	762	1262	30428		
1912	湖北	13	117.8	70	3474	2886	99064		
1912	湖南	14	102.9	32	1275	1437	52329		
1912	四川	15	131.9	203	6352	10661	345383		
1912	广东	16		9	715	1794	86437		
1912	广西	17		34	1222	1328	51097		
1912	云南	18	107.7	36	1544	1944	57808		
1912	贵州	19	101.9	5	130	1811	27036		
1912	福建	20	125.5	12	576	678	29653		
1912	甘肃	21	125.6	3	40	1243	22996		
1912	新疆	22	113	10	112	462	6910		

Figure 11: Census and Education Information for Different Provinces in 1909 and 1912
Source: Author extract from books and papers

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